CO3 - ATL 2

1. A Shepherd boy gets bored tending the towns flock. to have some fun he cries out, "wolf" even though no wolf in sight. The villagers run to protect the flock; but then get really mad when they realize the boy was playing a joke on them. One night, the shepherd boy sees a real wolf approaching the flock and calls out, "wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. the town goes hungry. panic ensures. Identify TP, FP, FN, TN.

Sol:

Actual/predicted	Y	N
Y	TP (real wolf came, warned)	FN
N	FP (no wolf, false warning)	TN

2. Assume there are 100 images, 30 of them depict a Cat, the rest do not. A machine learning model predicts the occurence of a Cat in 25 of 30 cat images. It also predicts absence of a Cat in 50 to 70 no Cat images. Identify TP, FP, FN, TN. and Calculate the precision and Recall and F1 score

Sol: Total images = 100

Actual Cat images = 30

Actual not Cat images = 70

Actual \ predicted	Actual cat Y	Actual No cat N
pre' Y	TP	FN
N	FP	TN

25	20	45
5	50	55
30	70	100

* precision $= \frac{TP}{TP+FP} = \frac{25}{25+5} = \frac{25}{30} = \frac{5}{6} = 83.3\%$

* Recall $= \frac{TP}{TP+FN} = \frac{25}{\cancel{70}} = \cancel{50\%}$ 35.714%

* F $= \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = \frac{2 \times 0.833 \times 0.357}{1.19}$

$= 0.5$

(3). An email service provider wants to classify emails as Spam or Not Spam using features:

* Contains "offer" (Yes/No)
* Contains "free" (Yes/No)
* Email length (short/long)

(1) Calculate prior probabilities (P(spam), P(Not Spam))

(2) Compute Conditional probabilities for each feature

(3) using naive Bayes, classify : (Offer = yes, free = No, length = short)

offer	Free	length	class
yes	yes	short	spam
yes	NO	long	notspam
NO	yes	short	spam
NO	NO	long	notspam
yes	yes	long	spam
NO	NO	short	notspam.

Sol: There are total 6 mails: spam - 3, notspam-3

① Calculate prior probabilities:

$$P(Spam) = \frac{\text{total spam mails}}{\text{total mails}} = \frac{3}{6} = 0.5$$

$$P(notspam) = \frac{\text{total notspam mails}}{\text{total mails}}$$

$$= \frac{3}{6} = 0.5 \text{ (or) } 1-0.5 = 0.5$$

② Calculate Conditional probabilities

we need to calculate the probability of every feature value given class.

$$P(\text{feature} | \text{class}) = \frac{\text{no. of occurences in feature in class}}{\text{total no. of emails in class}}$$

Ⓐ Conditional probability for spam:

offer	Free	length
yes	yes	short
NO	yes	short
yes	yes	long

total spam mails = 3

$P(\text{offer} = yes / spam) = \frac{2}{3}$ offer=NO occurs 1time : 1/3 free given spam: $P(\text{free} = \text{yes} | \text{spam}) = \frac{3}{3} = 1$

$P(\text{free} = \text{No} | \text{spam}) = 0$.

$P(\text{length} = \text{short} | \text{spam}) = 2/3$

$P(\text{length} = \text{long} | \text{spam}) = 1/3$

Spam Conditional probability table:

~~feature~~	value	probability
offer	yes	2/3
Offer	NO	1/3
free	yes	1
free	NO	0
length	short	2/3
length	long	1/3

(B) Conditional probability for not Spam:

$P(\text{offer} = \text{yes} | \text{Not spam}) = 1/3$

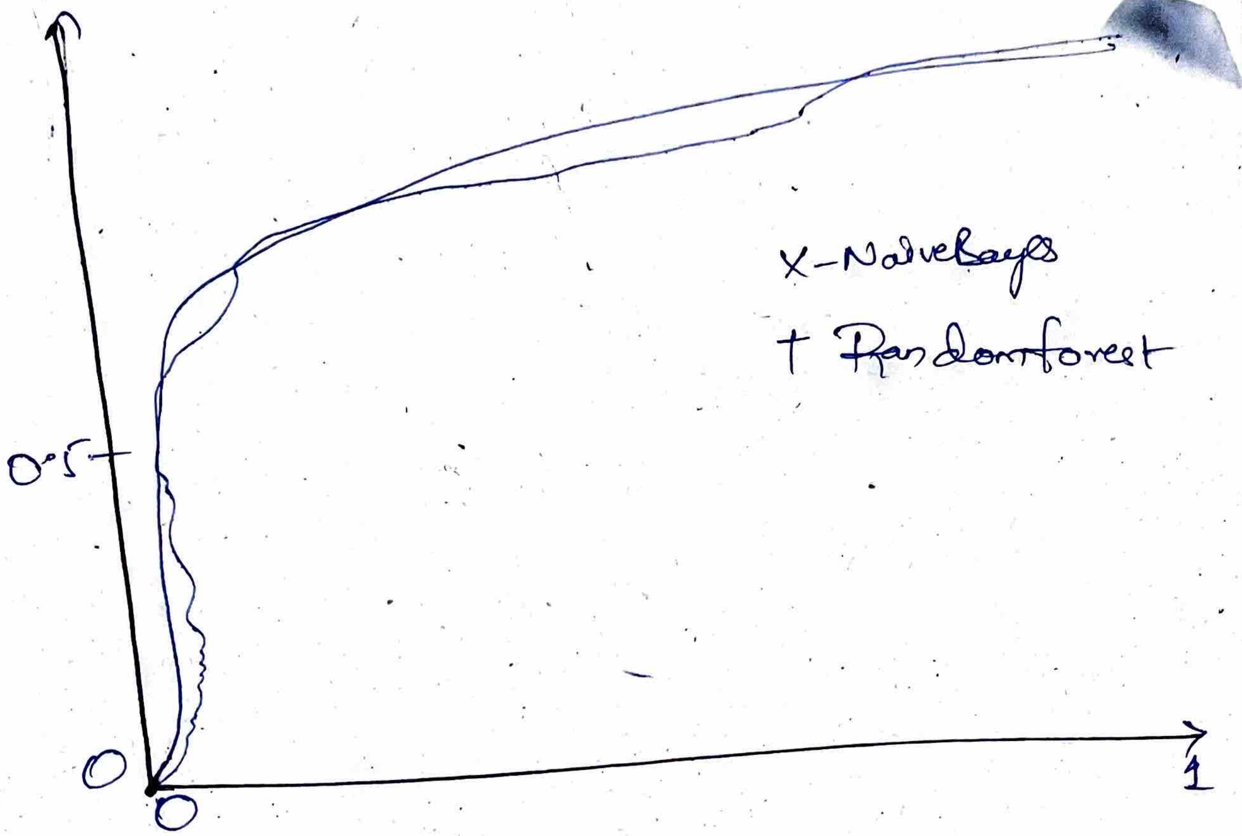
$P(\text{offer} = \text{No} | \text{Not spam}) = 2/3$

$P(\text{free} = \text{yes} | \text{Not spam}) = 0$

$P(\text{free} = \text{No} | \text{Not spam}) = 3/3 = 1$

$P(\text{length} = \text{short} | \text{Not spam}) = 1/3$

$P(\text{length} = \text{long} | \text{Not spam}) = 2/3$


X-NaiveBayes
+ Randomforest
0.5
0
0
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